

Ground Control Witness Form & Diagrams

Project Name: Chesterton Windmill, Windmill Hill Lane, Leamington Spa, CV33 9LB, UK

Date: 09/08/2025

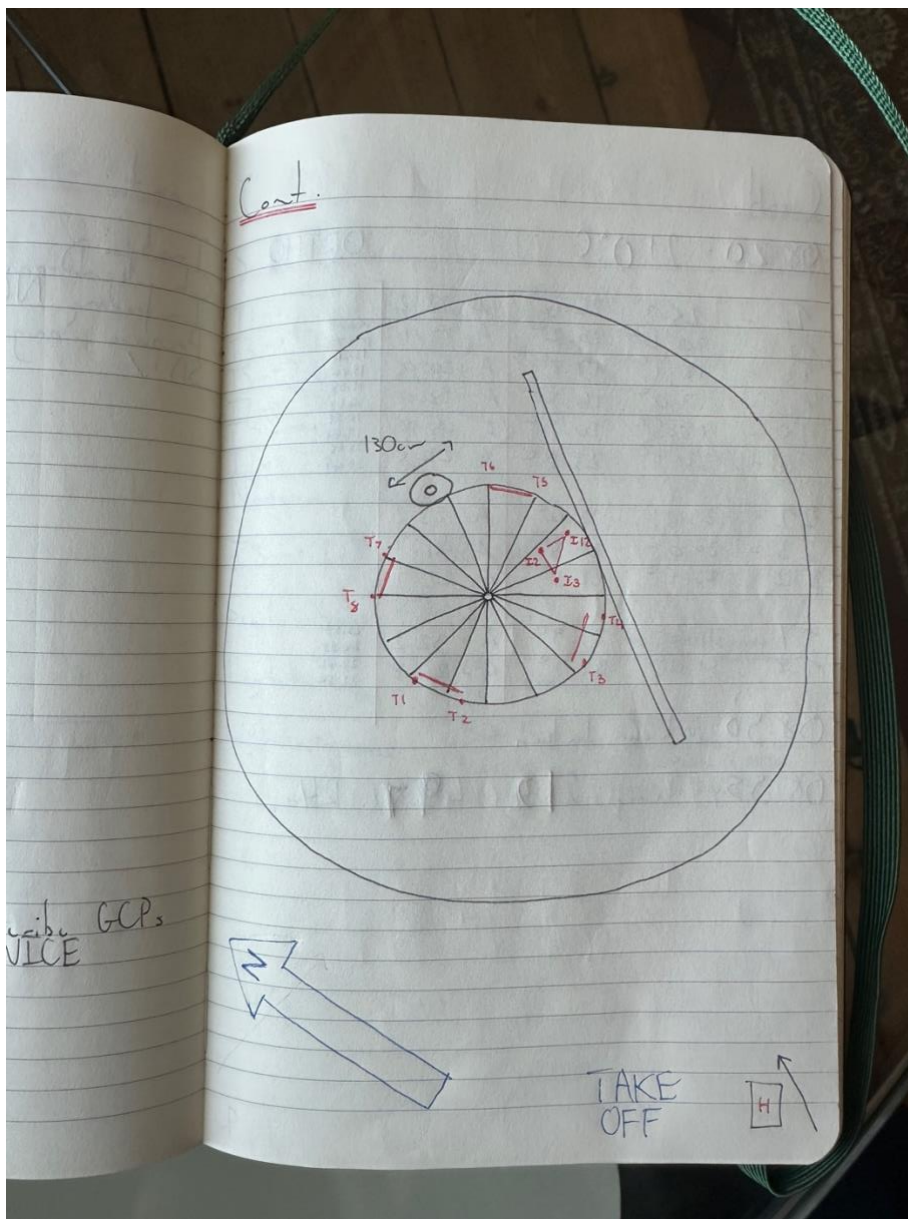
Project Code: WIND

	Details	Notes/Comments	Justification/Explanation
Survey Overview			
Date:	09/08/2025	Date of survey	
Survey Conditions:	Clear day. Survey undertaken at 08:40 whilst sun was coming up.	Brief description of survey conditions.	
No. of Points	9 SCALE BARS!!!! (7 using markers, 2 using natural features)	Number of points being taken.	NO INDIVIDUAL 3D POINT COORDINATES, only scale bars (due to listed building restrictions)
Equipment:			
Model:	N/A	Make/model of surveying equipment used. This includes the make/model of any TST, prism, tripod	
Specifications:	N/A	Critical specifications only inc. accuracy, range etc.	
Additional Equipment	Tape measure used by two surveyors between thermal-specific survey markers.	Equipment needed for the execution of the survey i.e. tape measure	
Survey Team:			
Planned / Set Up by:	Neil Sutherland	Lead individual responsible for survey planning	
Observed by:	Neil Sutherland Steve Sutherland	The individual taking the control point observations. This should remain consistent during the survey.	Measurement taken by both individuals to reduce human error
Computed by:	Neil Sutherland	The team member responsible for the processing of the observations	

Notes:	
Notes:	SCALE BARS: DUE TO THE NATURE OF THE SITE, SCALE BARS WERE TAKEN BETWEEN THERMAL-SPECIFIC SURVEY MARKERS. 7 'Thermal' Scale bars, denoted by markers starting with 'T...' 4 external scale bars around the base of the Windmill 3 internal scale bars across one internal column (denoted by Int...'). Only one scale bar was included on the interior of the structure due to the poor thermal contrast. This column (Int2, Int3 and Int12) received the most solar radiation during the survey. 2 'Natural' Scale bars taken on features visible in all modalities on the windmill (i.e., TIR, RGB and VIS sensors) This included two lightning wires on opposing sides of the windmill. 'Sail' denotes the lightning wire on the side where the sails of the Windmill are present (during this survey), whilst 'NonSail' denotes the opposite side ACROSS from the sails. An accuracy of 3mm was given for each scale bar. Measurements were checked twice by both surveyors to reduce human error in the measurement of the scale bars. Measurements were taken between the crosshairs of each marker, where aluminium and rubber elements create a +. IMPORTANT: As RGB images were captured on a different day to the TIR and VIS datasets, markers were not present during this survey. This means the only way to check the quality of the RGB is using the natural markers. If additional control is required, additional scale bars can be generated by re-visiting the site.

Diagrams

Site Diagram 1:



Site Diagram 2:



SCALE BAR 1

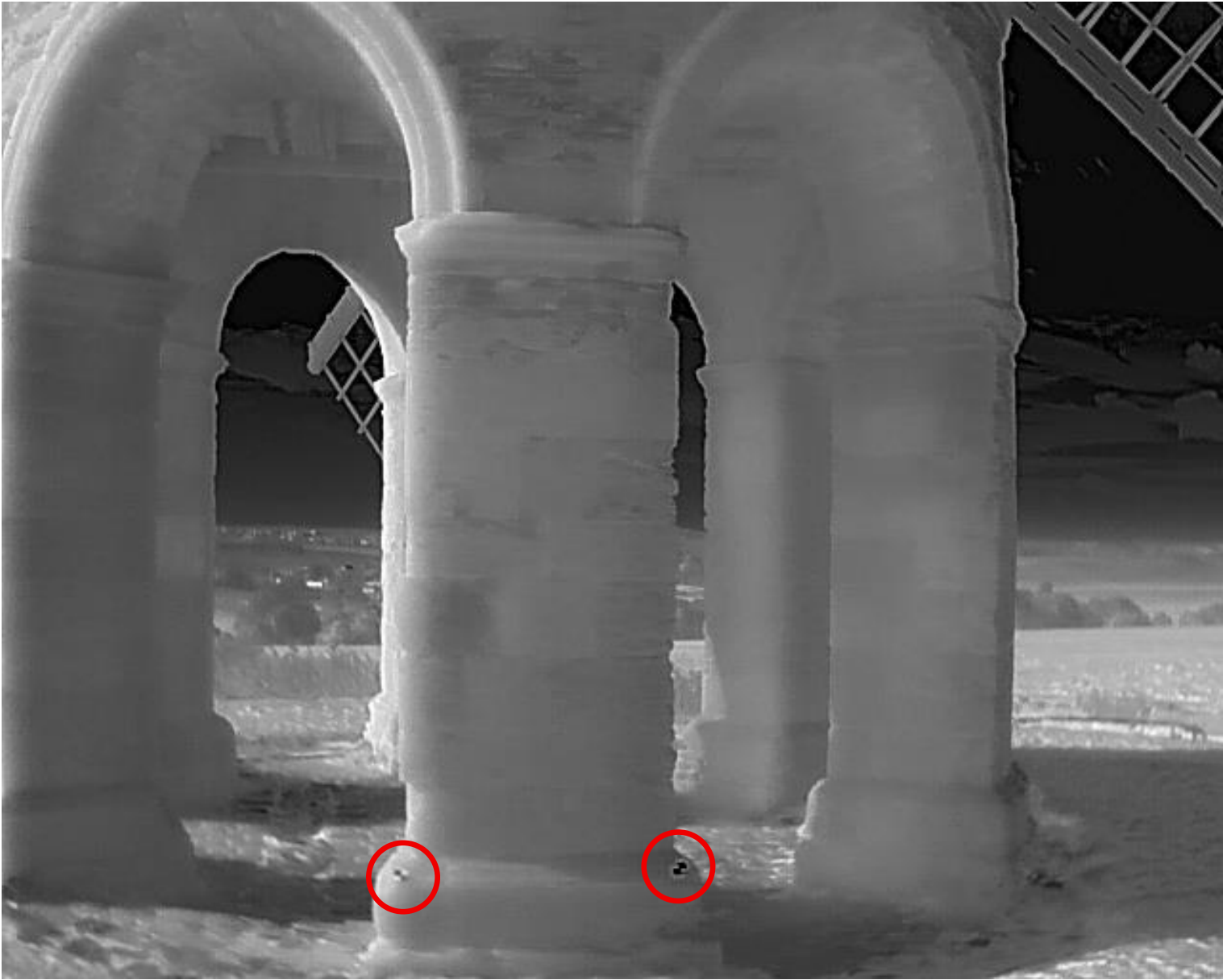
	Details	Notes/Comments	Justification/Explanation
Scale Bar	T1_T2	Name saved within the equipment	
Distance (m):	1.225		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	Scale bar taken between T1 and T2 survey markers.

Diagrams

**Control Point
Photo(s):**



Diagrams	
Control Point Diagram(s):	

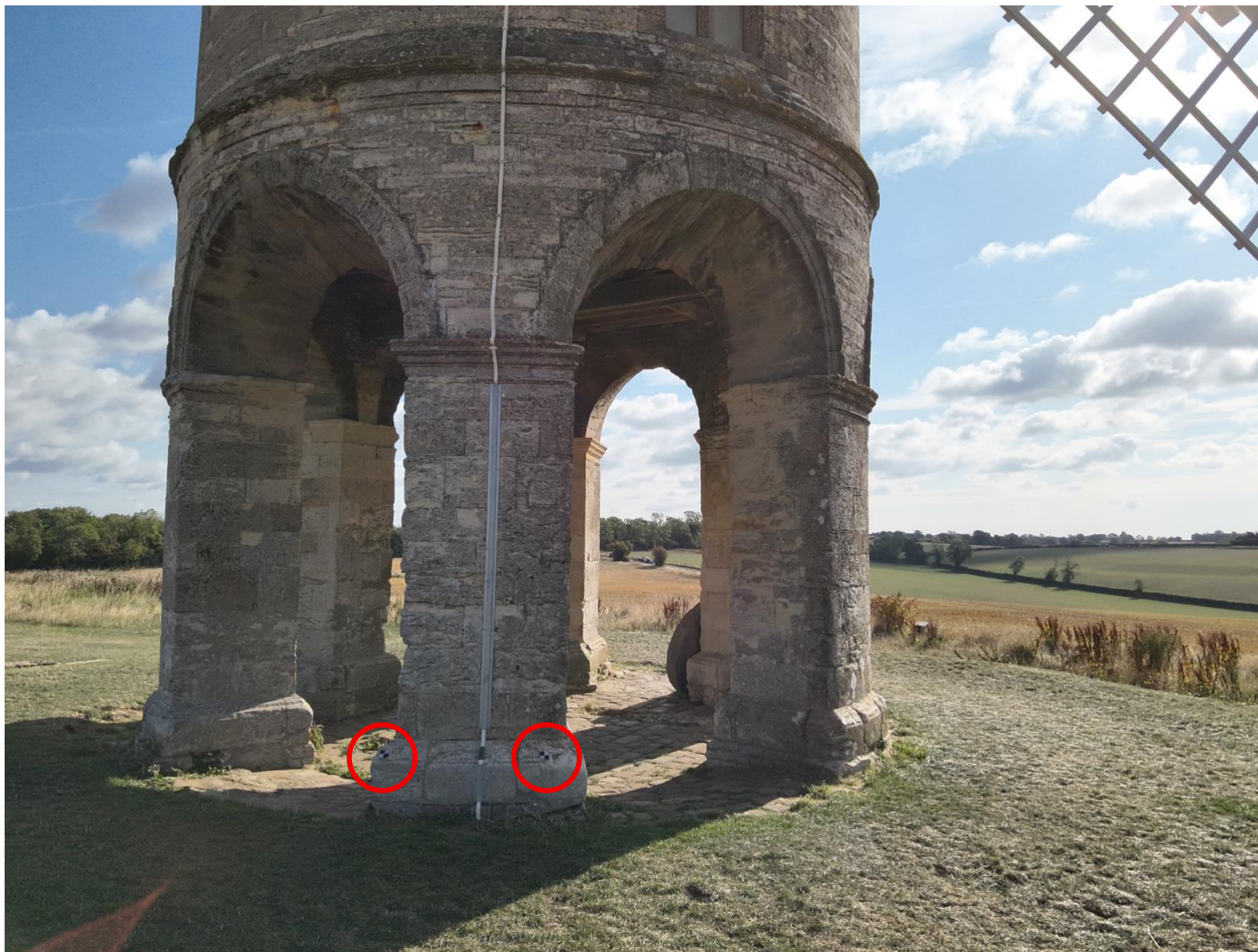
SCALE BAR 2

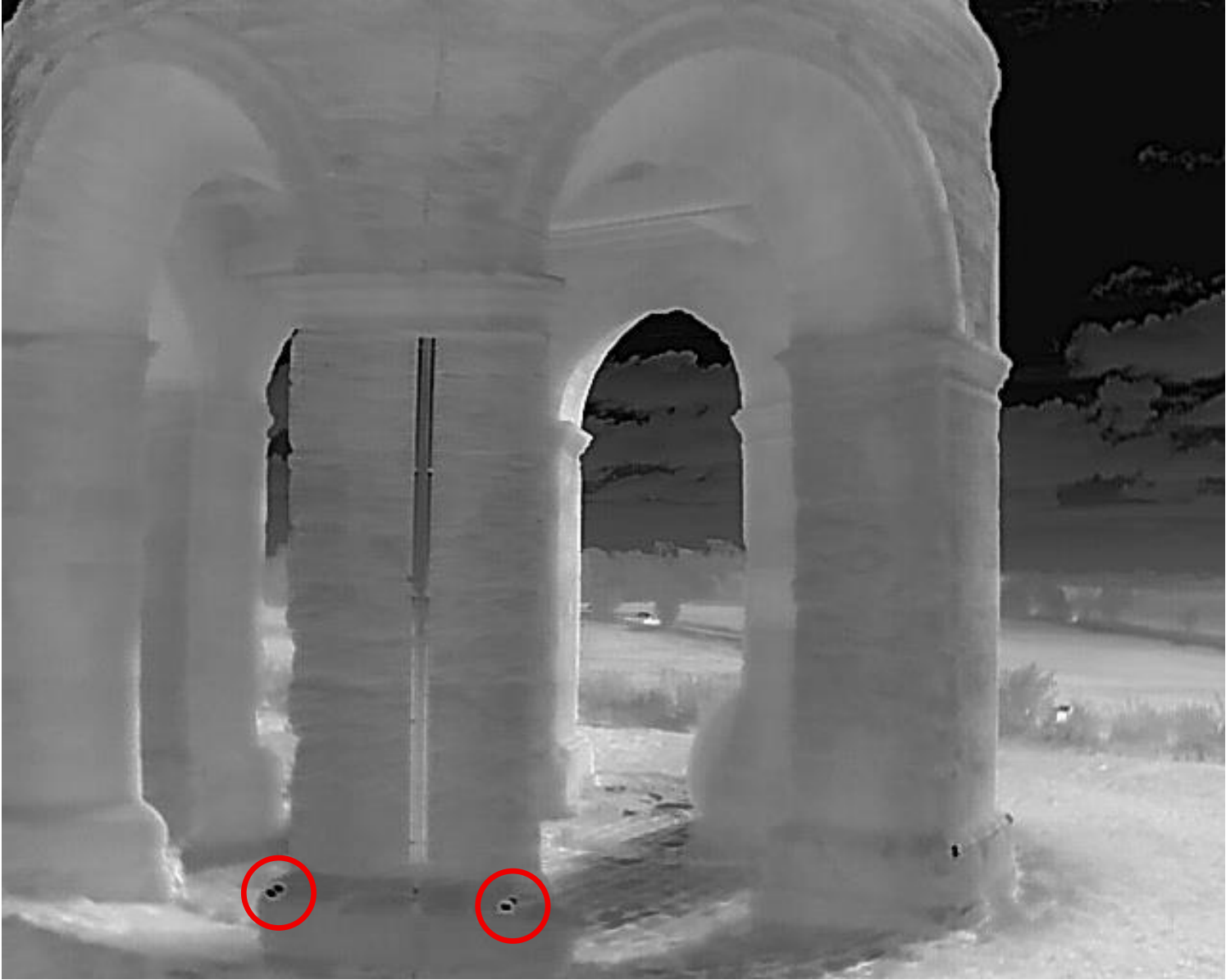
	Details	Notes/Comments	Justification/Explanation
Scale Bar	T3_T4	Name saved within the equipment	
Distance (m):	1.088		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	Scale bar taken between thermal-specific markers T3 and T4. This scale bar is underneath the sail of the windmill (during the survey) EITHER SIDE OF THE LIGHTNING WIRE for the Sail1_Sail2 scale bar.

Diagrams

**Control Point
Photo(s):**



Diagrams	
Control Point Diagram(s):	 A black and white photograph of a stone archway, likely part of a bridge or a large building. The archway is composed of several stone pillars and arches. Two red circles are drawn on the base of the pillars, highlighting specific control points. The background shows a landscape with trees and a body of water.

SCALE BAR 3

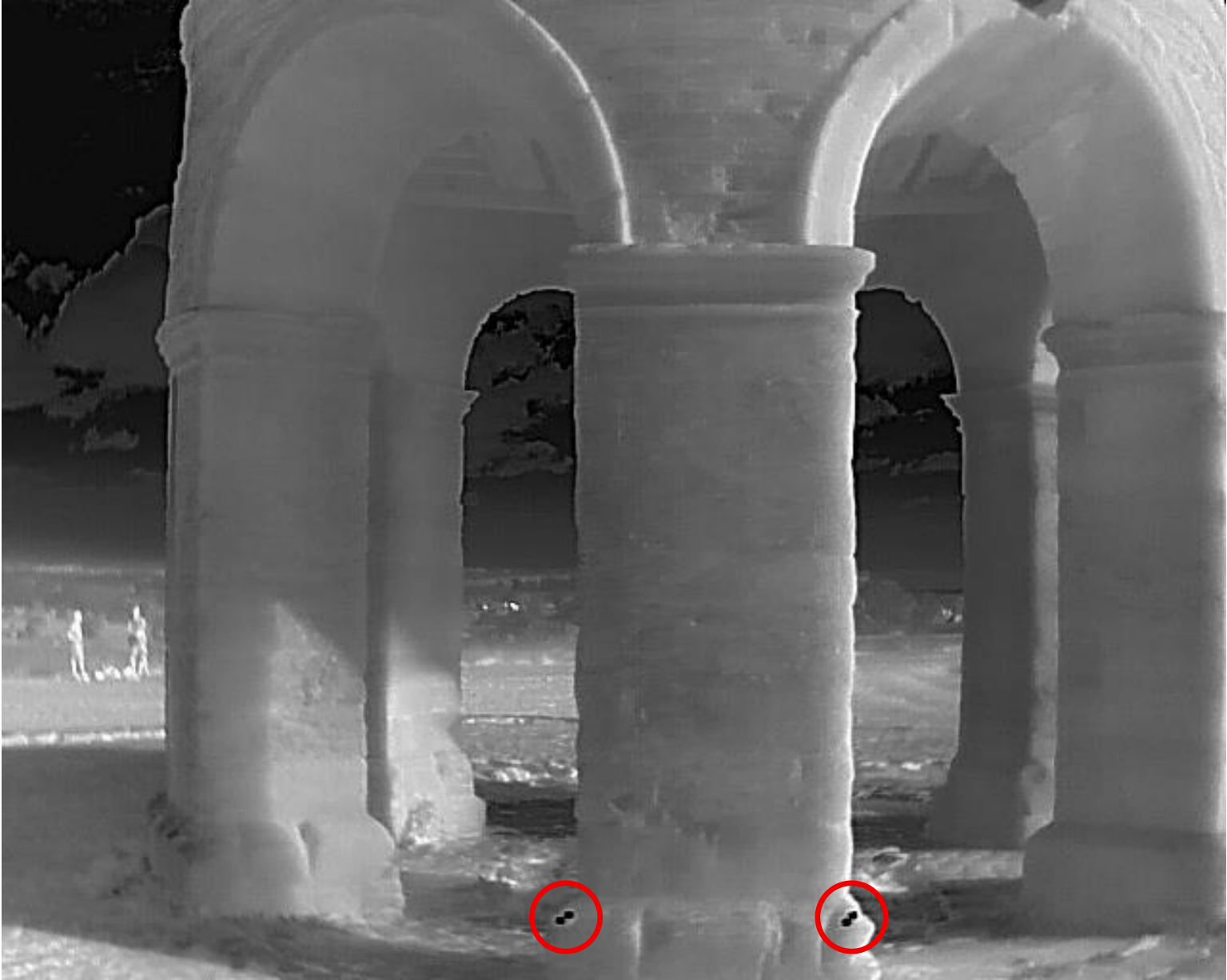
	Details	Notes/Comments	Justification/Explanation
Scale Bar	T5_T6	Name saved within the equipment	
Distance (m):	1.208		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	Scale bar taken between thermal-specific markers T5 and T6. These markers reside on the next column along from T3 and T4, whereas the others have a gap between them.

Diagrams

**Control Point
Photo(s):**



Diagrams	
Control Point Diagram(s):	 A black and white photograph of a stone archway, likely part of a historic building. The archway features a central column and two side columns, all supporting a series of arches. The ground is paved, and the background shows a grassy area with some trees. Two red circles are drawn on the image, highlighting specific control points at the base of the central column. The first circle is located at the bottom center of the column, and the second circle is located at the bottom right of the column. Both circles contain a small black dot, indicating the exact location of the control points.

SCALE BAR 4

	Details	Notes/Comments	Justification/Explanation
Scale Bar	T7_T8	Name saved within the equipment	
Distance (m):	1.252		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	Scale bar taken between thermal markers T7 and T8. These are either side of the lightning wire on the FAR SIDE from the sails, NonSail3 and NonSail4. These markers can be found just after the millstone (circular mill wheel).

Diagrams

**Control Point
Photo(s):**



Diagrams

**Control Point
Diagram(s):**



SCALE BAR 5

	Details	Notes/Comments	Justification/Explanation
Scale Bar	Tint2_Tint3	Name saved within the equipment	
Distance (m):	0.787		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	Internal markers visible on the column nearest the sails. Int2 represent the 'right-angle' of the triangle formed by Int2, Int3 and Int12

Diagrams

**Control Point
Photo(s):**



Diagrams	
Control Point Diagram(s):	 A grayscale photograph of a stone bridge over a river. The bridge has two large piers visible. Two red circles are drawn on the piers, each containing a small black crosshair, indicating control points for a survey or mapping project. The background shows a river with a small waterfall or dam, and a landscape with trees and buildings under a cloudy sky.

SCALE BAR 6

	Details	Notes/Comments	Justification/Explanation
Scale Bar	Tint3_Tint12	Name saved within the equipment	
Distance (m):	1.500		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	Internal markers visible on the column nearest the sails. Int2 represent the 'right-angle' of the triangle formed by Int2, Int3 and Int12. Int 3 and Int 12 represent the acute corners of the triangle.

Diagrams

**Control Point
Photo(s):**



Diagrams	
Control Point Diagram(s):	 A grayscale photograph showing a stone pillar in the foreground. The pillar is positioned in the center-right of the frame. Two red circles are drawn on the pillar to indicate control points: one is located on the upper part of the pillar, and the other is on the lower part, near the base. The background features a landscape with a body of water, a distant shoreline with trees and buildings, and a sky with clouds. The ground in the foreground is paved with cobblestones.

SCALE BAR 7

	Details	Notes/Comments	Justification/Explanation
Scale Bar	Tint2_Tint12	Name saved within the equipment	
Distance (m):	1.278		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	Internal markers visible on the column nearest the sails. Int2 represent the 'right-angle' of the triangle formed by Int2, Int3 and Int12. Int2 -12 represent the vertical scale bar between the thermal-specific markers.

Diagrams

**Control Point
Photo(s):**



Diagrams	
Control Point Diagram(s):	 A grayscale photograph showing a stone pillar in the foreground. Two control points are marked with red circles: one on the upper part of the pillar and one on the base. The background features a landscape with a body of water, trees, and a cloudy sky.

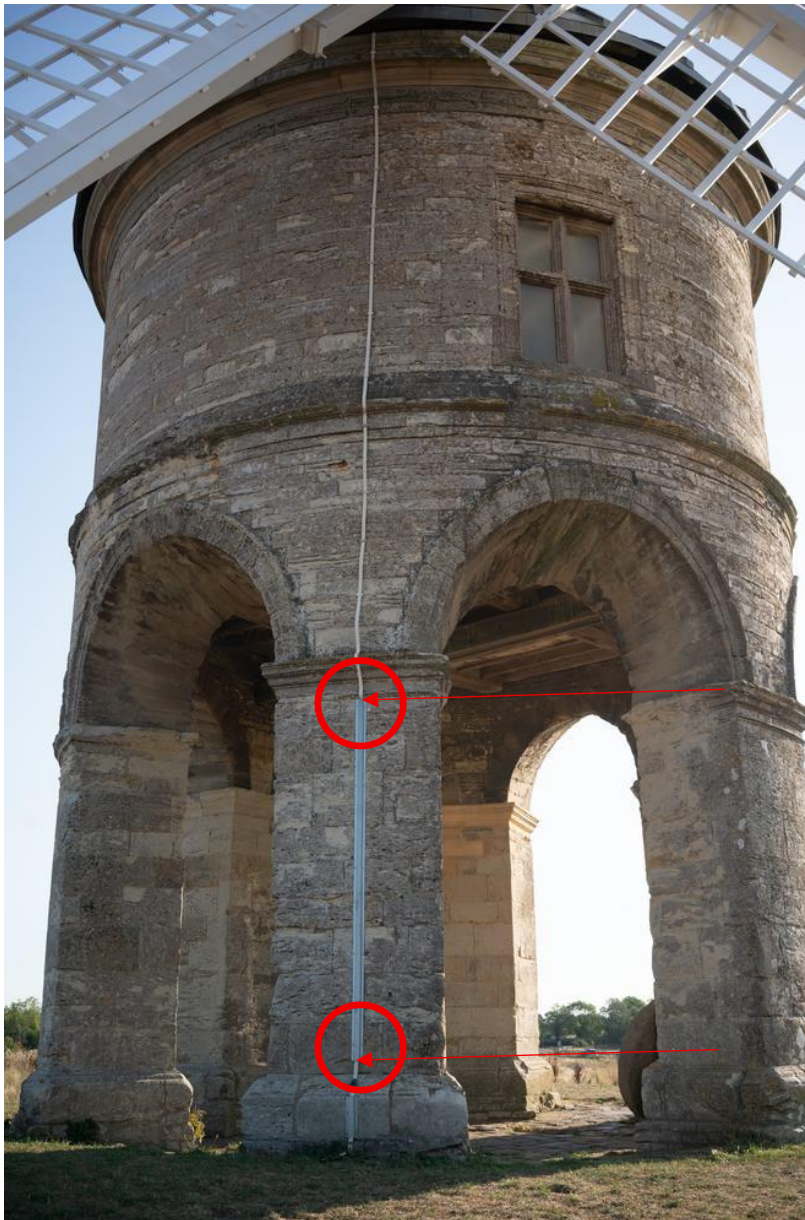
SCALE BAR 8

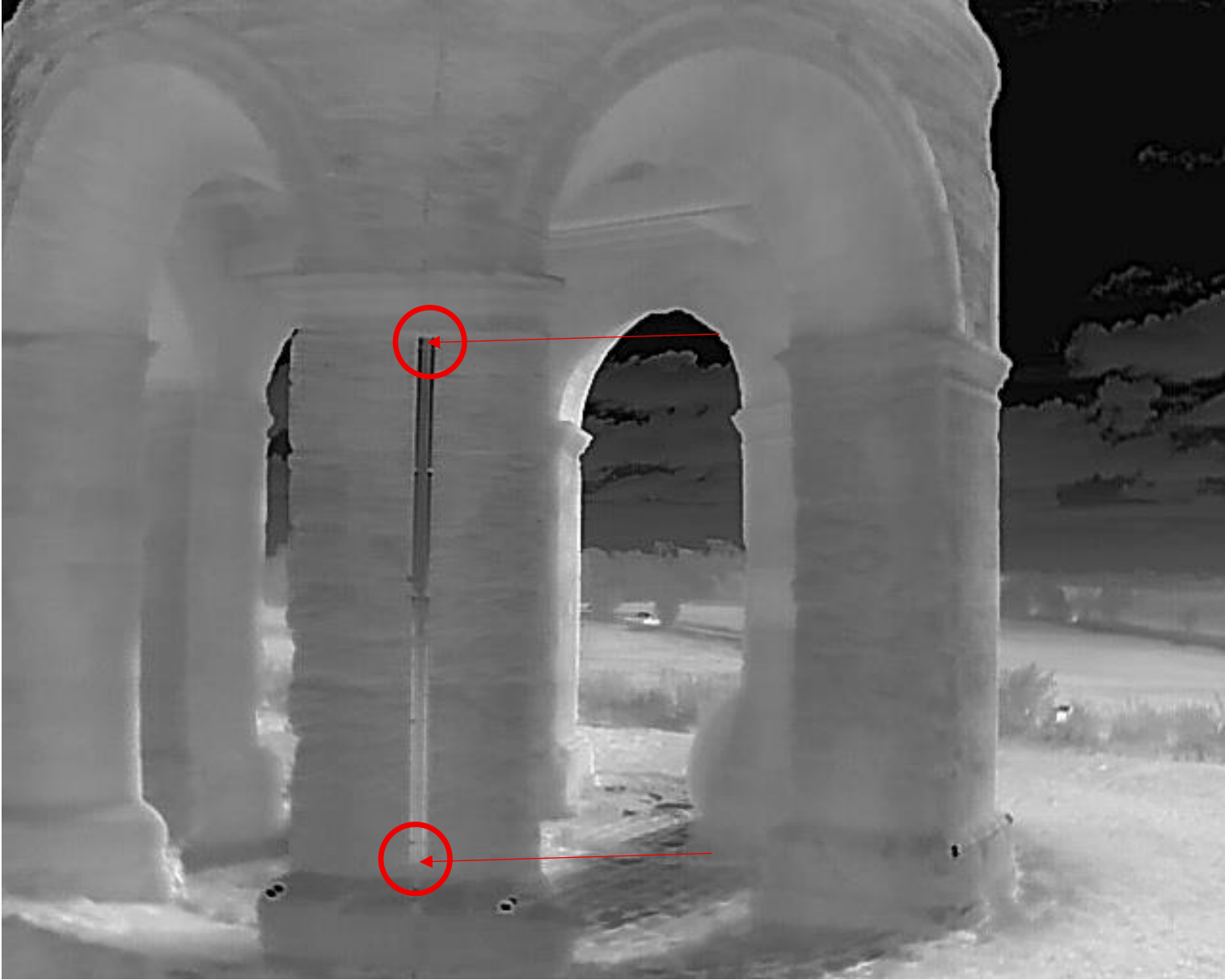
	Details	Notes/Comments	Justification/Explanation
Scale Bar	N_Sail1_N_Sail2	Name saved within the equipment	
Distance (m):	2.391		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	N_Sail 1 and N_Sail 2 are taken at the TOP and BOTTOM of the lightning wires metal casing, respectively. This is the main section of the metal casing, NOT down to the column plinth. This is the SAIL SIDE, underneath the windmill sails!

Diagrams

**Control Point
Photo(s):**



Diagrams Control Point Diagram(s):	
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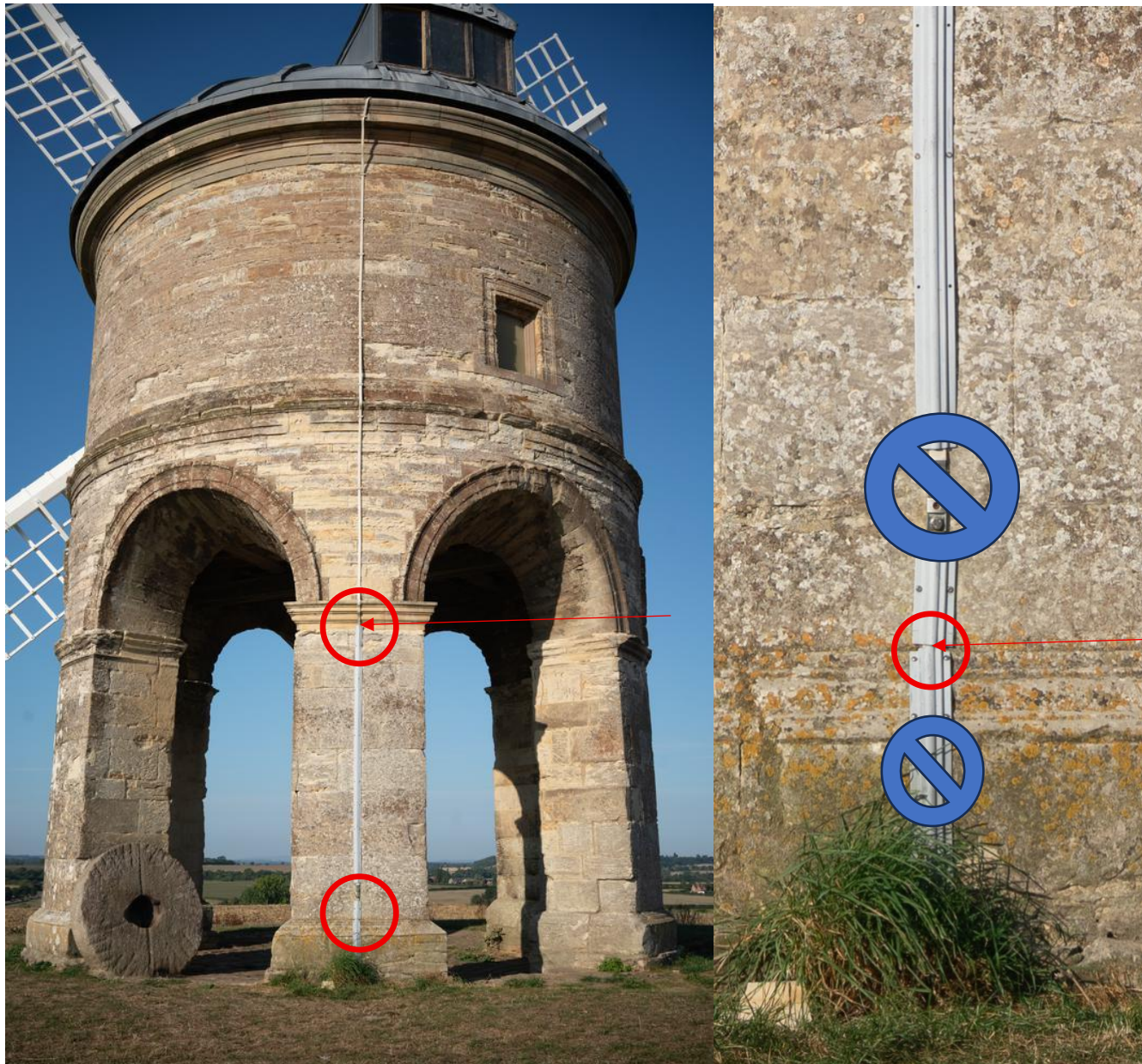
SCALE BAR 9

	Details	Notes/Comments	Justification/Explanation
Scale Bar	Non_Sail3_Non_Sail4	Name saved within the equipment	
Distance (m):	2.545		
Accuracy (m):	0.003		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	Non_Sail3 and Non_Sail4 are taken at the TOP and BOTTOM of the lightning wires metal casing, respectively. This is the main section of the metal casing, NOT down to the column plinth. Please refer to images below. This is the SAIL SIDE, underneath the windmill sails!

Diagrams

**Control Point
Photo(s):**



Diagrams

**Control Point
Diagram(s):**



Diagrams	
Additional Images:	